Computational aspects of penalized least-squares trend estimation with controlled smoothness

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Abstract

Penalized least-squares trend estimators provide an interpretable way to separate persistent variation from short-term fluctuations. Their practical use nevertheless requires choices about difference order, reference differences, smoothing intensity, and numerical implementation. We examine these choices through a reproducible computational framework for finite-difference penalties. We derive the polynomial null space and the spectral representation of the smoother, and normalize the trace-based smoothness index so that its range is independent of the difference order. We also study the conditional profile estimator of a nonzero reference-difference mean, distinguishing this derivation from procedures available in the software. Controlled experiments compare naive and cached dense linear systems, a reusable singular-value decomposition, and a banded solver. We separately investigate repeated full-trend reconstruction and scalar effective-degrees-of-freedom evaluations, including numerical agreement and preprocessing time. A nonlinear synthetic example illustrates the behavior of classical smoothing criteria, while contemporaneous NOAA atmospheric CO_2 and not seasonally adjusted US effective federal funds rate observations illustrate alternative selection criteria, difference orders, sampling windows, and explicitly nonprobabilistic trend extrapolation. The results highlight workload-dependent computational trade-offs rather than the universal superiority of any individual algorithm. The accompanying Python implementation and executable experiments provide a transparent basis for further numerical and statistical investigation.

Keywords: Whittaker–Henderson smoothing, spectral decomposition, smoothing parameter selection, time-series trend, reproducible computation

1 Introduction

An observed time series may combine low-frequency behavior with temporary movements and measurement noise. Estimating its trend is a descriptive task: the resulting curve depends on how fidelity to the observations is traded against smoothness. Whittaker's graduation framework and subsequent finite-difference penalized least-squares (PLS) estimators formalize this compromise Whittaker (1923); Guerrero (2007). The Hodrick–Prescott filter is an important second-difference special case, but the statistical problem admits other penalty orders and nonzero reference values for the differences Cortés-Toto et al. (2017).

The tuning parameter must also be chosen. Cortés-Toto, Guerrero, and Reyes Cortés-Toto et al. (2017) investigate attained smoothness when choosing the penalty with cross-validation (CV), generalized cross-validation (GCV), corrected Akaike information criterion (AICc), or Bayesian information criterion (BIC). Their simulations vary trend shape, series length, variability, and seasonality. Efficient Whittaker–Henderson computation has an extensive history, including methods exploiting banded factorization and structure to reduce numerical expense Weinert (2007); consequently, diagonalization is not presented here as a newly discovered mathematical algorithm.

In the scientific-computing literature, Balam et al. Balam et al. (2025) compare normal equations, QR, and singular-value decomposition (SVD) in least-squares computation and discuss conditioning. Azofeifa and Moreles Azofeifa and Moreles (2025) present a different numerical model, but similarly distinguish the formulation, solver, numerical comparisons, and limitations. We similarly distinguish statistical formulation, numerical algorithms, empirical comparisons, and their limitations.

Our objective is to provide a small, auditable Python package, `trendestimation`, that implements the *published* PLS family and exposes examples as executable notebooks within Sphinx documentation. Specifically, we describe the affine solution for general order d and reference mean μ, its spectral implementation, the criteria available in the public API, and numerical experiments that report computational time, solver agreement, and precomputation costs alongside reproducible illustrations. Experimental forecast-horizon cross-validation and dynamic tracking of minima from other research projects are not included. The distinction between an in-sample smoother and an ex ante forecast is maintained throughout.

2 Model and spectral representation

2.1 Finite-difference penalized least squares

Let $y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ denote an equally spaced series. For an integer $1 \le d < n$, let $D_d \in \mathbb{R}^{(n-d)\times n}$ represent the dth forward-difference operator; for example, row entries of D_2 are $(1, -2, 1)$. We consider the published estimator

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y - \tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}, \qquad \lambda \ge 0, \tag{1}$$

where $\mu \in \mathbb{R}$ is a fixed reference mean for the differences. The choice $\mu = 0$ gives the familiar pure penalty. The implementation additionally permits $d = 0$, with $D_0 = I_n$, as a well-defined edge case; the geometric statements below concern $1 \le d < n$.

Write $Q_d = D_d^\top D_d$ and choose $p \in \mathbb{R}^n$ satisfying $D_d p = \mu \mathbf{1}$. Using zero-based indices $t = 0, \dots, n-1$, one convenient choice is $p_t = \mu \binom{t}{d}$ because $D_d \binom{t}{d} = 1$. Setting $x = \tau - p$ and differentiating the strictly convex objective gives

$$\widehat{\tau}_{\lambda,\mu} = p + H_\lambda (y - p), \qquad H_\lambda = (I_n + \lambda Q_d)^{-1}. \tag{2}$$

Indeed, $(I_n + \lambda Q_d)x = y - p$ and $I_n + \lambda Q_d$ is positive definite for every finite $\lambda \ge 0$. This affine expression is equivalent to the normal equations $(I_n + \lambda Q_d)\widehat{\tau} = y + \lambda \mu D_d^\top \mathbf{1}$.

2.2 Proof of the *d*-dimensional kernel

For $1 \le d < n$, the dth forward difference has the explicit binomial stencil

$$(D_d \tau)_t = \sum_{k=0}^{d} (-1)^{d-k} \binom{d}{k} \tau_{t+k}, \quad t = 0, \dots, n-d-1. \tag{3}$$

The first $n-d$ columns of D_d form an upper triangular $(n-d) \times (n-d)$ submatrix with nonzero diagonal $(-1)^d$. Hence $\operatorname{rank}(D_d) = n - d$. As $Q_d = D_d^\top D_d$, $x^\top Q_d x = \|D_d x\|^2$, and therefore

$$\ker Q_d = \ker D_d, \qquad \dim \ker Q_d = n - \operatorname{rank}(D_d) = d. \tag{4}$$

The vectors $1, t, \dots, t^{d-1}$ have vanishing dth forward differences and are linearly independent on $n > d$ nodes. Thus they give a basis of the kernel, proving that exactly d eigenvalues of Q_d vanish in exact arithmetic. Under $\mu = 0$, the limit $\lambda \to \infty$ is the least-squares projection onto polynomials of degree $\le d - 1$. For instance $d = 1$ preserves a constant, $d = 2$ a line, and $d = 3$ a quadratic. When $\mu \ne 0$, the limiting set is instead the affine space $\{\tau : D_d \tau = \mu \mathbf{1}\}$, which has the same d homogeneous directions but includes a degree-d particular solution. In the implementation $p_t = \mu \binom{t}{d}$.

2.3 Interpretation of higher orders and the spline connection

For a sufficiently smooth function f observed on an equally spaced mesh of width h, Taylor expansion yields $\Delta_h^d f(x_t) = h^d f^{(d)}(x_t) + O(h^{d+1})$. Consequently, under appropriate mesh-refinement assumptions,

$$h^{1-2d} \sum_t (\Delta_h^d f(x_t))^2 \longrightarrow \int |f^{(d)}(x)|^2 \, dx. \tag{5}$$

This is a mathematical bridge to roughness penalties in smoothing splines. A Whittaker-type trend smoother penalizes differences of fitted ordinates directly, while a P-spline instead penalizes differences of coefficients in a chosen spline basis; these

are related ideas but not identical estimators. For a continuous penalty weight λ_{cont}, the corresponding discrete coefficient is scaled approximately as $\lambda_{\text{disc}} = \lambda_{\text{cont}} h^{1-2d}$.

The order d is an integer representing the order of the derivative being approximated; it is not a continuous interpolation between derivative orders. Comparing different d values is a change of regularization model, so equal numerical λ does not imply equal smoothing. It is useful to examine orders $d = 1, \ldots, 5$ under a common normalized trace-based index; however, arbitrary orders and polynomial nullspaces have long been known Weinert (2007); Biessy (2026). Larger d can be ill-conditioned because the binomial stencil has growing coefficients and the positive eigenvalues may span many orders of magnitude. Neither higher d nor higher polynomial degree guarantees a better trend or numerical stability.

2.4 Null space, SVD, and limiting trends

The rows of D_d are independent in exact arithmetic, so $\operatorname{rank}(D_d) = n - d$ and $\dim \ker D_d = d$. The null space consists of the sampled polynomials of degree at most $d-1$. Let $D_d = U \Sigma V^\top$ be a full SVD. With nonnegative eigenvalues δ_j of Q_d, ordered so that exactly d are zero, the smoother becomes

$$H_\lambda = V \operatorname{diag}\left(\frac{1}{1+\lambda\delta_1}, \ldots, \frac{1}{1+\lambda\delta_n} \right) V^\top. \tag{6}$$

Therefore the transformed observations $c = V^\top (y - p)$ can be computed once, and every new penalty value changes only the scalar attenuation factors $(1 + \lambda\delta_j)^{-1}$. At $\lambda = \infty$ the exact limiting fit is $p + P_{\ker D_d}(y - p)$, rather than a computation using an arbitrarily large finite penalty.

For fixed μ, the effective degrees of freedom are $\operatorname{EDF}(\lambda) = \operatorname{tr}(H_\lambda) = \sum_j (1 + \lambda\delta_j)^{-1}$. We report Guerrero's trace-based smoothness $S_G = 1 - \operatorname{tr}(H_\lambda)/n$ alongside the normalized coordinate

$$S(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n - d} \in [0, 1], \quad S(0) = 0, \quad S(\infty) = 1. \tag{7}$$

For $1 \le d < n$, this coordinate is strictly increasing for finite λ because $S'(\lambda) = (n-d)^{-1} \sum_{\delta_j > 0} \delta_j / (1 + \lambda\delta_j)^2 > 0$. The normalization changes the scale of the original index but not the ordering of smoothness levels Guerrero (2008).

3 Classical parameter selection and numerical implementation

3.1 Selection criteria

For a fixed numerical value of μ, define $e_\lambda = y - \widehat{\tau}_{\lambda,\mu}$, $\operatorname{RSS}_\lambda = \|e_\lambda\|_2^2$, $h_{ii} = (H_\lambda)_{ii}$, and $\nu_\lambda = \operatorname{tr}(H_\lambda)$. The published classical selectors studied in Cortés-Toto et al. (2017)

are represented, up to constants irrelevant to an optimizer, by

$$\mathrm{CV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_{\lambda,i}}{1-h_{ii}}\right)^2, \tag{8}$$

$$\mathrm{GCV}(\lambda) = \frac{\mathrm{RSS}_\lambda/n}{(1-\nu_\lambda/n)^2}, \tag{9}$$

$$\mathrm{AICc}(\lambda) = \log(\mathrm{RSS}_\lambda/n) + \frac{2\nu_\lambda+1}{n-\nu_\lambda-2}, \tag{10}$$

$$\mathrm{BIC}(\lambda) = \log(\mathrm{RSS}_\lambda/n) + \frac{\nu_\lambda \log n}{n}. \tag{11}$$

The software checks denominators and excludes undefined candidates, evaluates a declared positive grid of penalty values, and compares an exact infinite-penalty candidate. In particular, a near-interpolating BIC minimum can be driven by the log(RSS) boundary, as our example illustrates. We do not interpret a finite-grid BIC minimum close to zero as a universally reliable statistical choice.

Guerrero Guerrero (2007) also considers the feasible plug-in $\widehat{\mu} = \mathrm{mean}(D_d y)$. This option is available *for fitting*; however, the formula for leave-one-out CV in (8) presumes that μ remains fixed under deletion. The current public selectors intentionally require a fixed numerical μ. Likewise, $\mathrm{tr}(H_\lambda)$ is a conditional EDF and is not automatically the full degrees of freedom of a procedure that estimates μ from the same observations.

3.2 Smoothing correction, inversion, and exact endpoints

With the full orthogonal eigenbasis of the penalty, $Q_d = V\,\mathrm{diag}(\delta_1,\ldots,\delta_n)V^\top$, the $n-d$ positive eigenvalues are $\delta_j = \sigma_j(D_d)^2$, and the remaining d are zero. Consequently, the spectral eigenvalues of $H_\lambda = (I+\lambda Q_d)^{-1}$ are

$$\alpha_j(\lambda) = \begin{cases} (1+\lambda\delta_j)^{-1}, & 1 \le j \le n-d, \\ 1, & n-d < j \le n. \end{cases} \tag{12}$$

This exposes both the exact limiting polynomial directions and the scalar attenuation of the penalized directions. If $\nu(\lambda) = \mathrm{tr}(H_\lambda) = d + \sum_{j=1}^{n-d}\alpha_j(\lambda)$, the original trace-based Guerrero index

$$S_G(\lambda) = 1 - \frac{\nu(\lambda)}{n} \quad \text{ranges from } 0 \text{ to } 1-\frac{d}{n}. \tag{13}$$

The upper bound is $1-d/n$, *not* $1-n/d$. The library also uses a normalized correction:

$$S(\lambda) = \frac{S_G(\lambda)}{1-d/n} = \frac{n-\nu(\lambda)}{n-d} \in [0,1]. \tag{14}$$

The map is strictly increasing for $\lambda \in [0, \infty)$ because

$$S'(\lambda) = \frac{1}{n-d} \sum_{j=1}^{n-d} \frac{\delta_j}{(1+\lambda\delta_j)^2} > 0. \tag{15}$$

The exact endpoints are $S(0) = 0$ and $S(\infty) = 1$; for any interior target, a one-dimensional bracketed solver inverts this map. Equal normalized smoothness makes trace scales comparable across d; it does not make the underlying trend geometries or roughness penalties identical.

3.3 Where the computational acceleration comes from

For fixed μ put $p_t = \mu\binom{t}{d}$ and $c = V^\top(y - p)$. One common projection computes c; after that, for each candidate λ,

$$\operatorname{tr}(H_\lambda) = d + \sum_{j=1}^{n-d} \alpha_j(\lambda), \tag{16}$$

$$\|y - \widehat{\tau}_{\lambda,\mu}\|^2 = \sum_{j=1}^{n-d} (1 - \alpha_j(\lambda))^2 c_j^2. \tag{17}$$

Thus a sweep of M values can obtain trace, normalized smoothness, GCV, AICc, BIC and related scalar profiles in $O(Mn)$ *after* the shared eigendecomposition and $O(n^2)$ projection. A full reconstructed trend remains $O(n^2)$ per penalty, and exact ordinary CV leverage diagnostics also involve the spectral basis. The full dense SVD has $O(n^3)$ preprocessing time and $O(n^2)$ storage. This decomposition/reuse trade-off explains the potential responsiveness of repeated-criteria explorations without claiming unconditional superiority to banded linear solvers Weinert (2007); Biessy (2026).

Large batches of spectral attenuation operations are naturally vectorizable. Dense linear algebra can also benefit from optimized multithreaded BLAS on CPUs. In principle, a GPU backend could parallelize matrix decompositions and batched matrix operations, but host/device transfer, synchronization, and initialization costs may negate a benefit for short series. We found `numpy` and `scipy.linalg.svd` in the package currently reviewed, not a released GPU solver; therefore no GPU speedup, power saving, or measured device execution is reported. A future benchmark should record CPU and GPU models, precision, BLAS and accelerator versions, transfer time, warmups, memory, and the number of candidate penalties.

3.4 Reusable computation and cost

The package exposes a reusable `TrendSmoother(n, order=d)` object. Construction forms the difference operator and its SVD. For each candidate λ, a fitted curve is obtained with the transformed vector c and spectral attenuation in (6). The library offers `fit`, `degrees_of_freedom`, `smoothness`, and `select_lambda`, while plotting

utilities visualize fits and criterion profiles. Sphinx incorporates executable Jupyter notebooks, with deterministic random seeds and figure code for inspection.

The computational trade-off must distinguish decomposition from reuse. A dense full SVD costs $O(n^3)$ preprocessing and $O(n^2)$ memory; after projection, EDF, smoothness, GCV, AICc, and BIC summaries cost $O(n)$ per penalty candidate, while full reconstruction and exact CV leverage scores require $O(n^2)$ each. In contrast, the banded structure of $I_n + \lambda Q_d$ allows efficient specialized direct solvers Weinert (2007). Thus the phrase *fast computation* means repeated spectral evaluation under a stated workload, not a blanket claim that SVD beats banded factorization. A separate reproducible benchmark in the repository measures setup and fit times independently.

4 Gaussian marginal likelihood and profiling the difference mean

4.1 Likelihood selection for fixed μ

A probabilistic interpretation of the penalty assumes observation noise $\varepsilon \sim N(0, \sigma^2 I_n)$ and stochastic differences $D_d\tau = \mu\mathbf{1}+\eta$ with $\eta \sim N(0, \sigma^2\lambda^{-1}I_{n-d})$. The d unpenalized polynomial coordinates act as nuisance fixed effects. In the $r = n - d$ positive-penalty singular coordinates $z_j = v_j^\top y$, write $p_t^{(1)} = \binom{t}{d}$, $b_j = v_j^\top p^{(1)}$. Marginalization gives

$$z_j \mid \mu, \lambda, \sigma^2 \sim N\left(b_j\mu, \sigma^2 c_j(\lambda)\right), \quad c_j(\lambda) = 1 + \frac{1}{\lambda\delta_j}. \tag{18}$$

This is a Gaussian residual/marginal-likelihood interpretation, not a new form of PLS. It is related to likelihood-based Whittaker–Henderson smoothing Biessy (2026). Profiling the variance σ^2 for a fixed numerical μ leaves, up to an additive constant,

$$J(\lambda; \mu) = r \log\left(\frac{1}{r}\sum_{j=1}^{r}\frac{(z_j - b_j\mu)^2}{c_j(\lambda)}\right) + \sum_{j=1}^{r}\log c_j(\lambda). \tag{19}$$

An independently developed branch of the package, `feature/mle-smoothness-and-data-mu-20261021`, adds this profiled Gaussian score for fixed μ or the conditional plug-in $\mu = \overline{D_d y}$. That plug-in must not be mislabeled as a joint MLE.

4.2 A closed-form estimator of μ conditional on λ

Let $w_j(\lambda) = c_j(\lambda)^{-1} = \lambda\delta_j/(1 + \lambda\delta_j)$. For a fixed positive λ, the generalized least-squares optimum satisfies

$$\widehat{\mu}(\lambda) = \frac{\sum_{j=1}^{r} w_j b_j z_j}{\sum_{j=1}^{r} w_j b_j^2}. \tag{20}$$

The denominator is positive because $D_d p^{(1)} = \mathbf{1}$. Equation (20) is also obtained by minimizing the PLS objective jointly in τ and μ for a fixed λ. Substitution in (19)

Table 1 Selected penalty and normalized smoothness for the case in Fig. 1. Trend RMSE is computed against the known latent simulation only.

Criterion	$\widehat{\lambda}$	S	Trend RMSE
CV	6.17	0.777	0.242
GCV	6.14	0.777	0.242
AICC	9.74	0.805	0.248
BIC	2.45e-07	0.000	0.454

gives the one-dimensional concentrated score $J(\lambda; \widehat{\mu}(\lambda))$, whose optimization produces a joint profile-likelihood candidate $(\widehat{\lambda}, \widehat{\mu})$. Such a solution depends on the Gaussian variance model, may lie near a boundary, and must be checked for identifiability and other local minima. This paper provides the equation and a separately reproducible numerical test; joint μ profiling is not yet an exposed release API.

In fact, taking the limit $\lambda \downarrow 0$ using $w_j \sim \lambda\delta_j$ proves

$$\lim_{\lambda\downarrow 0} \widehat{\mu}(\lambda) = \frac{(p^{(1)})^\top Q_d y}{(p^{(1)})^\top Q_d p^{(1)}} = \frac{\mathbf{1}^\top D_d y}{n-d} = \overline{D_d y}. \tag{21}$$

For positive λ the two expressions differ in general. Estimating μ adds a rank-one adjustment to the data-to-trend linear operator, so the fixed-μ effective degrees of freedom and ordinary leave-one-out denominators are not automatically valid after joint estimation. Prior literature already considers nonzero difference means Guerrero (2007); Cortés-Toto et al. (2017); the implementation contribution here is an explicit, testable profiling workflow, not a claim to have first introduced μ or likelihood selection.

5 Reproducible numerical illustration

5.1 Source-inspired nonlinear design

To demonstrate the interface without introducing an external data source, we reproduce the nonlinear simulation family of Cortés-Toto et al. Cortés-Toto et al. (2017):

$$\tau(t) = 0.6 f_{\text{Beta}(30,17)}(t/n) + 0.4 f_{\text{Beta}(3,11)}(t/n), \quad t = 1, \ldots, n. \tag{22}$$

We take $n = 50$, add independent normal errors with standard deviation 0.5, omit seasonality, and fix the pseudo-random seed at 19. These choices define one illustrative replication, not a full reproduction of the article's factorial simulation. We fit $d = 2$, $\mu = 0$, and select λ from the same 121-point geometric grid augmented with the exact $\lambda = \infty$ candidate under each criterion. The latent trend in (22) is known, allowing an *ex post* trend-recovery RMSE; it is not a future forecast error.

As shown in Table 1, CV and GCV agree in this particular realization, whereas AICc selects a somewhat smoother fit. BIC selects a penalty extremely close to zero on the inspected finite grid; its apparent interpolation tendency illustrates a known

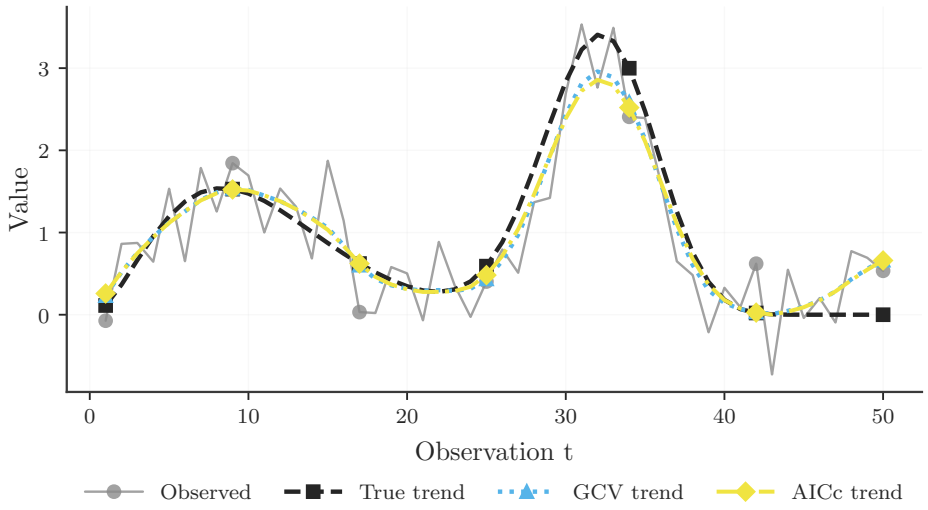


Fig. 1 A synthetic nonlinear case based on Cortés-Toto et al. Cortés-Toto et al. (2017). The curves show *in-sample* estimates and known simulated truth, not out-of-sample forecasts. Source data are regenerated by `figures/make_figures.py`.

practical boundary concern and is not evidence that BIC is generally ineffective. The simulated recoveries are descriptive; no general method ranking follows from a single seed.

5.2 Algebraic and numerical checks

An independent check compares the SVD implementation with the direct normal equations for $n = 50$, $\lambda = 30$, $d \in \{1, 2, 3\}$ and $\mu \in \{0, 0.15\}$ on a second fixed-seed Gaussian sample. The maximum componentwise absolute difference is 2.01×10^{-12} across these six cases, and the limiting constraint $D_d \widehat{\tau}_{\infty,\mu} = \mu \mathbf{1}$ holds to maximum error 4.42×10^{-12}. Both values result from the supplied deterministic regression script in IEEE double precision; they verify the tested algebra rather than establish stability for all n, d, λ. Higher differencing orders can be ill-conditioned, and systematic machine-dependent speed measurements remain a distinct stage of the study.

6 Measured computational cost and repeated parameter evaluation

6.1 Experimental design and independent baselines

Numerical correctness and elapsed time are distinct properties. We therefore benchmark the same PLS estimator at matched (n, d, μ) and identical candidate penalties, using four implementations: (i) a deliberately naive dense solver that reconstructs D_d,

Table 2 Leading arithmetic complexity for M reconstructed trends of length n and fixed order d. Setup is counted and no exact operation counts or GPU measurements are asserted.

Method	Setup	Repeated full fits
Fresh dense solves	rebuilt per λ	$O(Mn^3)$
Cached dense Q_d	dense penalty matrix	$O(Mn^3)$
Spectral SVD	$O(n^3)$	$O(Mn^2)$
Direct banded solver	$O(nd^2)$	$O(Mnd^2)$

Table 3 Host-measured end-to-end median times in milliseconds, including one-time setup, for M full fitted trends. Every method computes the same trends for the same n, $d = 2$ and $\mu = 0$; R is the naive-fresh-dense time divided by the spectral time. Values and the numerical parity check are generated from the raw CSV files, not entered manually.

n	M	Fresh	Cached	Spectral	Banded	R
40	1	0.029	0.027	0.204	0.123	0.14
40	64	1.388	0.739	1.302	1.032	1.07
100	1	0.075	0.073	0.801	0.281	0.09
100	64	4.589	2.642	2.013	1.532	2.28
200	1	0.471	0.350	3.580	0.535	0.13
200	64	20.984	10.348	4.982	2.017	4.21
400	1	3.192	3.217	20.165	1.103	0.16
400	64	196.554	106.894	20.885	3.188	9.41

$Q_d = D_d^\top D_d$, and solves the normal equations at every candidate; (ii) a denser but fairer baseline that caches Q_d and repeats its dense solve; (iii) the reusable full-SVD spectral implementation of `TrendSmoother`; and (iv) a banded solver that assembles Q_d directly in its sparse-band storage and uses a structure-aware linear solver. The last comparison is crucial: beating dense normal equations does not imply beating specialized PLS solvers Weinert (2007).

For every case a deterministic Gaussian sequence is generated, μ is held at either zero or a fixed nonzero value, and λ spans 10^{-3} to 10^3 on a geometric grid. The `quick` experiment varies $n \in \{40, 100, 180\}$, $M \in \{1, 16, 64\}$ with $d = 2$ and adds $d \in \{1, 3, 5\}$ at $n = 100$, $M = 24$, together with a nonzero-μ case. The `full` experiment reaches $n = 400$, $M = 256$ and additional difference orders. All results include one-time setup (not only post-SVD solves); we record medians, extrema, and median absolute deviations over repeated independent runs. We bound maximum disagreement against an independent banded solve for *every* tested penalty. Runs use a declared random seed, source revision, machine metadata, NumPy/SciPy versions and a stated BLAS-thread limit. The code and machine-readable outputs are confined to `paper/benchmarks/`.

Table 4 Median end-to-end times (milliseconds) to evaluate M effective-degrees-of-freedom traces, including setup. These entries measure only scalar $\mathrm{tr}(H_\lambda)$ quantities, not reconstructed trends.

n	M	Fresh	Cached	Spectral	Banded
40	8	0.213	0.150	0.203	0.322
100	8	1.059	0.832	0.801	1.072
100	32	4.220	3.307	0.885	3.615
180	32	15.914	12.208	2.830	10.114
250	64	92.614	59.421	6.090	38.391

6.2 End-to-end full-fit timing results

The crossover between methods should be evaluated using *end-to-end* times, including the SVD setup cost, for both a single smoothing parameter and a repeated-λ sweep. A ratio greater than one denotes a speed advantage over the naive baseline on the reported machine, not a universal speedup. In particular, the structured banded baseline can outperform a reusable SVD for full trend reconstruction.

6.3 Scalar smoothness and EDF are different workloads

To evaluate $\mathrm{tr}(H_\lambda)$ repeatedly, the spectral method needs only $n - d$ positive eigenvalues:

$$\mathrm{tr}(H_\lambda) = d + \sum_{j=1}^{n-d} (1 + \lambda\delta_j)^{-1}.$$

This is $O(Mn)$ work *after* spectral setup, not $O(Mn^2)$ reconstructed trends. An independent second experiment compares repeated dense computations of $\mathrm{tr}[(I + \lambda Q_d)^{-1}]$, a banded solver applied to all n canonical right-hand sides, and the spectral sum. The experiment checks algebraic equivalence separately from timed regions. Even when the spectral trace is the most efficient *marginal* score after setup, the $O(n^3)$ SVD cost can still dominate small workloads.

Measurement provenance: Platform: Windows-10-10.0.26100-SP0; CPU: Intel64 Family 6 Model 183 Stepping 1, GenuineIntel; Logical CPUs: 32; NumPy: 2.4.6; SciPy: 1.17.1; BLAS threads: 1; Repetitions: 7; Preset: full; Revision: local-unrecorded.
Maximum observed absolute trend disagreement: $4.32e-10$; maximum EDF trace disagreement: $5.48e-12$.

6.4 Interpreting speed and computational cost

A timed speed ratio is defined by the two end-to-end medians, for example $R_{\mathrm{dense/spectral}} = t_{\mathrm{dense}}/t_{\mathrm{spectral}}$. A value above one favors the spectral approach for the tested workload, not for arbitrary data lengths or hardware. CPU time, wall time, arithmetic complexity and energy consumption are different quantities. CPU numerical libraries already use optimized native linear algebra, but this experiment does

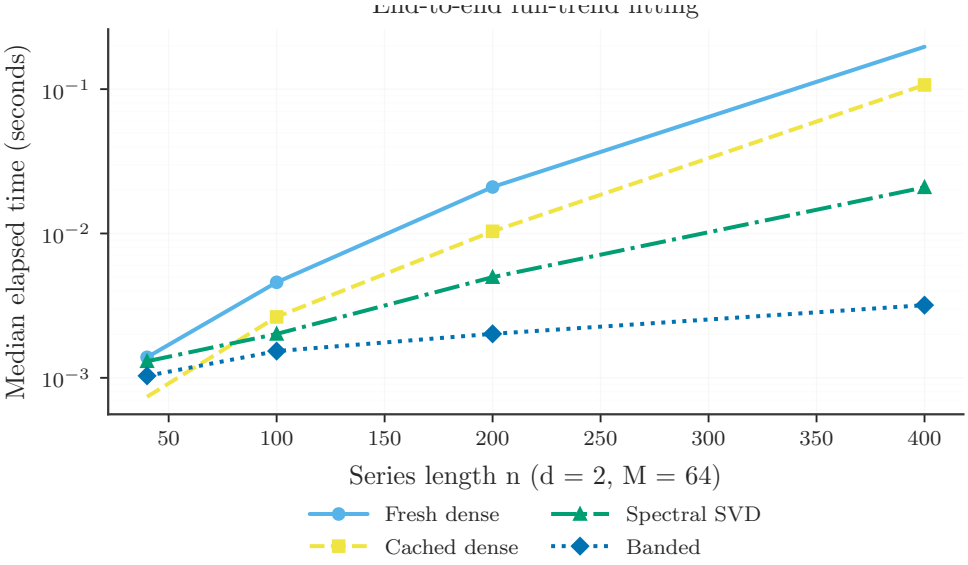


Fig. 2 End-to-end median times for $M = 64$ fitted curves, including setup, generated on the executing host from the raw CSV files. These are hardware-specific measurements, not an assumption that spectral reconstruction is the universally fastest method.

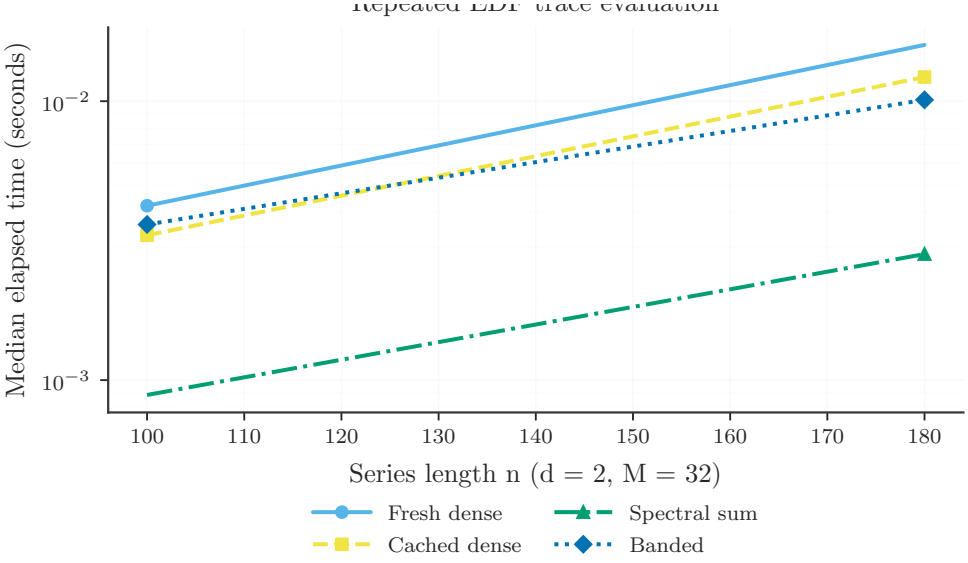


Fig. 3 Repeated effective-degrees-of-freedom trace evaluations ($M = 32$), including preprocessing. The quantity is scalar $\mathrm{tr}(H_\lambda)$, not a full trend reconstruction.

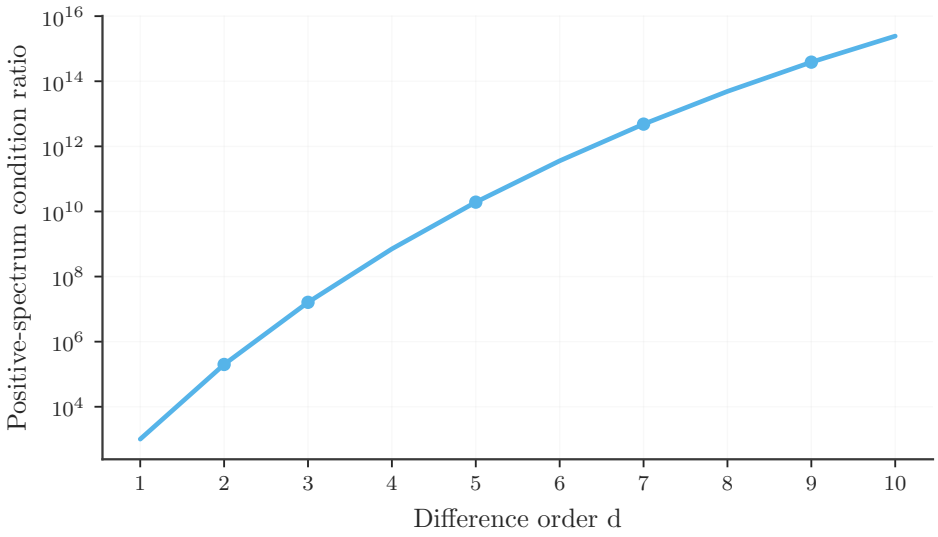


Fig. 4 Condition ratio of the positive spectrum of $D_d^\top D_d$ for $n = 50$, $d = 1, \ldots, 10$. Values are recomputed from singular values of D_d; they do not represent hardware-specific runtime.

not establish GPU execution or electricity use. An explicit device and power measurement would be necessary to make either claim. Results are intentionally generated only through a manual command, so the manuscript does not inadvertently contain timings from a different machine presented as those of its authors.

7 A controlled exploration of larger difference orders

The software accepts any integer $0 \leq d < n$, but this algebraic admissibility must be distinguished from floating-point stability. We therefore conduct a deterministic diagnostic with $n = 50$, seed 24021, difference orders $d = 1, \ldots, 10$, $\mu = 0$, and common normalized smoothness $S = 0.86$. At each order, we form the full SVD of D_d, compute the condition ratio

$$\kappa_+(Q_d) = \frac{\max_{1 \leq j \leq n-d} \delta_j}{\min_{1 \leq j \leq n-d} \delta_j}, \tag{23}$$

and compare the spectral fitted trend with an independent direct solution of $(I + \lambda Q_d)\tau = y$. The ratio is calculated from the *strictly positive* penalty eigenvalues; the full Q_d has zero eigenvalues and is singular for every $d \geq 1$.

The ratio rises from about 10^3 at $d = 1$ to 2.44×10^{15} at $d = 10$. At $d = 10$, the maximum componentwise difference between the SVD-based fit and the direct normal-equation solution is approximately 3.36×10^{-7} for the declared sample and smoothing target, versus approximately 1.14×10^{-15} at $d = 1$. The direct normal-equation result is not presumed an exact ground truth at large condition numbers: both approaches are affected by finite precision. The experiment illustrates why support

for many difference orders needs accompanying conditioning diagnostics, rather than establishing new high-order theory, unconditional stability, or performance superiority. Machine-specific setup/fitting times are separately exported by the experiment script for later audit; they are not promoted to general speed benchmarks.

8 Contemporary observed-data case studies

8.1 Data sources and scientific scope

We replace the archived 1990–1999 carbon-dioxide demonstration in the submission with a frozen 441-month sequence of directly published NOAA Global Monitoring Laboratory (GML) Mauna Loa monthly-mean atmospheric CO_2 concentrations, January 1990–September 2026, in parts per million NOAA Global Monitoring Laboratory and Scripps Institution of Oceanography (2026). The September 2026 observation is 426.07 ppm; October 2026 is not included because a complete monthly value had not been released at the October 26 data cutoff. The source snapshot and provenance accompany the submission. Unlike the earlier `statsmodels` historical example, these are published monthly observations rather than calendar-month averages constructed from an archived weekly file. The strong seasonal cycle remains in the data; it should not be conflated with observational error or removed implicitly by PLS. We also study the most recent 120-month subset of this record to isolate the consequences of a shorter estimation window.

For an independent economic example, we use the US effective federal funds rate (FEDFUNDS), the monthly average of daily rates (not seasonally adjusted), from January 2010 through September 2026 ($n = 201$). The underlying source is the Board of Governors of the Federal Reserve System and the distribution is through the Federal Reserve Economic Data (FRED) service Board of Governors of the Federal Reserve System (2026). The September 2026 release reports 3.75 percent. The snapshot is retained with its download URL, last observation, checksum, and retrieval procedure. Both records may subsequently be revised by their publishers.

8.2 Four selectors, two parameterizations, four orders

For each series and $d \in \{1, 2, 3, 4\}$ we fix $\mu = 0$ and compare CV, GCV, AICc, BIC, and the controlled normalized smoothness $S = 0.86$. For the four score-based selectors we use a common 29-point interior smoothness grid $\mathcal{S} = \{0.10, 0.10 + 0.88/28, \ldots, 0.98\}$. Each candidate s is mapped to its unique penalty $\lambda = S^{-1}(s)$, and all four criteria are evaluated on the *same* resulting penalty candidates. Thus differences in the selected fitted curves are attributable to the criterion at a given d, rather than differing numerical search domains. The controlled-S fit is an explicit user choice, not a fifth automatically optimized criterion.

The experiment records both $\widehat{\lambda}$ and $\widehat{S} = S(\widehat{\lambda})$ for every selected fit. These are two coordinates for one smoother, not two independent estimators. When comparing d values, a common S equalizes the fraction of penalizable trace degrees of freedom, not the space of admissible polynomial components. In particular $\dim \ker(D_d) = d$, and higher orders may have less stable endpoint behavior.

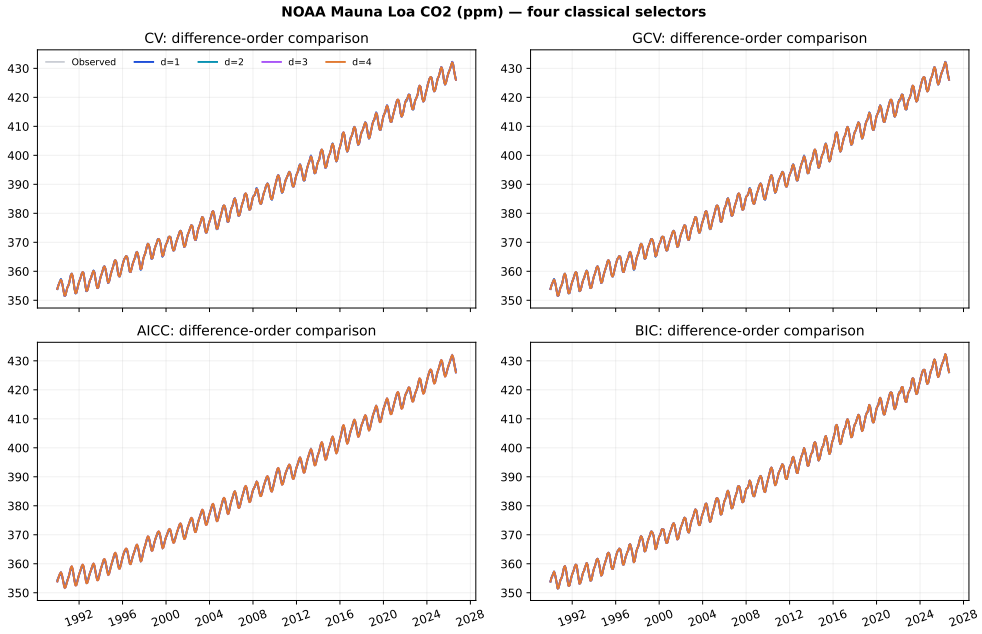


Fig. 5 Four classical criterion choices applied to NOAA Mauna Loa monthly mean CO_2, 1990–September 2026, at difference orders $d = 1, 2, 3, 4$ and fixed $\mu = 0$. All panels use identical candidate-S grids within each order. These are in-sample two-sided trends, not forecasts.

8.3 Future extension versus validated forecasting

The fitted trend is only defined at observed time indices. To illustrate possible trajectories *without attributing a forecasting model to the PLS smoother*, we regress the final 36 fitted values on a locally centered polynomial of degree $\min(d-1, 2)$ and evaluate that polynomial for the next 12 months, October 2026–September 2027. For $d = 1$ this is a constant continuation, $d = 2$ a linear continuation, and $d \geq 3$ a quadratic continuation. The degree cap is an explicitly imposed stability choice: an unrestricted degree-$d-1$ extrapolation for large d can diverge rapidly. These lines extrapolate the *smoothed trend only*; they do not forecast CO_2 seasonal components, policy shocks, or interest-rate policy changes, and no predictive intervals are inferred.

As an independent *single-origin diagnostic*, we additionally stop estimation 12 months before the published end date, choose the criterion and compute the fit using that training segment alone, and compare 12-step extrapolations with the held-out subsequent observations. This avoids look-ahead leakage from the test period. It is not a substitute for multiple rolling forecast origins, uncertainty calibration, or explicit seasonal models.

The principal purpose of these comparisons is to expose differences across *selection methods, penalty orders, sample periods and extrapolation conventions*, not to rank future predictive ability from one observed realization. A subsequent forecasting study should compare multiple chronological origins and horizons, explicitly account

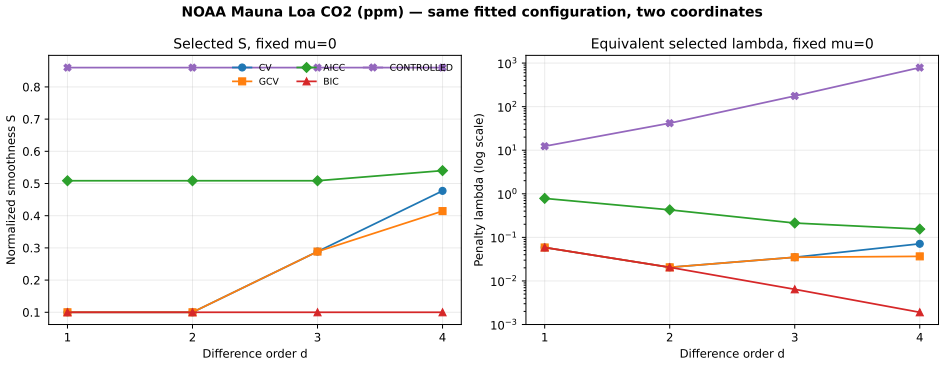


Fig. 6 The identical CO_2 selections reported as normalized smoothness $S(\widehat{\lambda})$ (left) and penalty $\widehat{\lambda}$ (right). The controlled value $S = 0.86$ is the prescribed point across orders; the other points minimize their named scores over the candidate grid.

Table 5 Reproducibly selected penalties, normalized smoothness and 12-month single-origin holdout RMSE for $d = 2$ across the full CO_2 record, its last 120 months, and not monthly effective federal funds rate. The controlled smoothness $S = 0.86$ is prescribed. CO_2 errors are in ppm and federal funds rate errors in percentage points; the RMSE values are *not* directly comparable across datasets. The 12-month evaluation is exploratory, not a rolling-origin performance estimate.

Dataset	Selector	λ	S	holdout RMSE
CO_2 (1990–2026)	CV	0.02055	0.100	2.597
CO_2 (1990–2026)	GCV	0.02055	0.100	2.597
CO_2 (1990–2026)	AICC	0.4284	0.509	2.596
CO_2 (1990–2026)	BIC	0.02055	0.100	2.597
CO_2 (1990–2026)	CONTROLLED	41.77	0.860	2.520
CO_2 (recent 120 months)	CV	0.02053	0.100	2.597
CO_2 (recent 120 months)	GCV	0.02053	0.100	2.597
CO_2 (recent 120 months)	AICC	0.4234	0.509	2.596
CO_2 (recent 120 months)	BIC	0.02053	0.100	2.597
CO_2 (recent 120 months)	CONTROLLED	37.03	0.860	2.524
Fed funds (2010–2026)	CV	0.02054	0.100	1.009
Fed funds (2010–2026)	GCV	0.02054	0.100	1.009
Fed funds (2010–2026)	AICC	1.212	0.634	1.012
Fed funds (2010–2026)	BIC	0.02054	0.100	1.009
Fed funds (2010–2026)	CONTROLLED	39.57	0.860	1.067

for seasonality and publication vintages, and report paired loss differences with uncertainty.

9 Conclusions

We have presented an auditable implementation of established penalized least-squares smoothing, together with additional derivations of the difference nullspace, the corrected smoothness index, and a Gaussian mean-difference profile estimator. The latter has numerical checks but is not yet a released joint-MLE API. Our current NOAA CO_2 and US effective federal funds rate observations provide real-data illustrations of

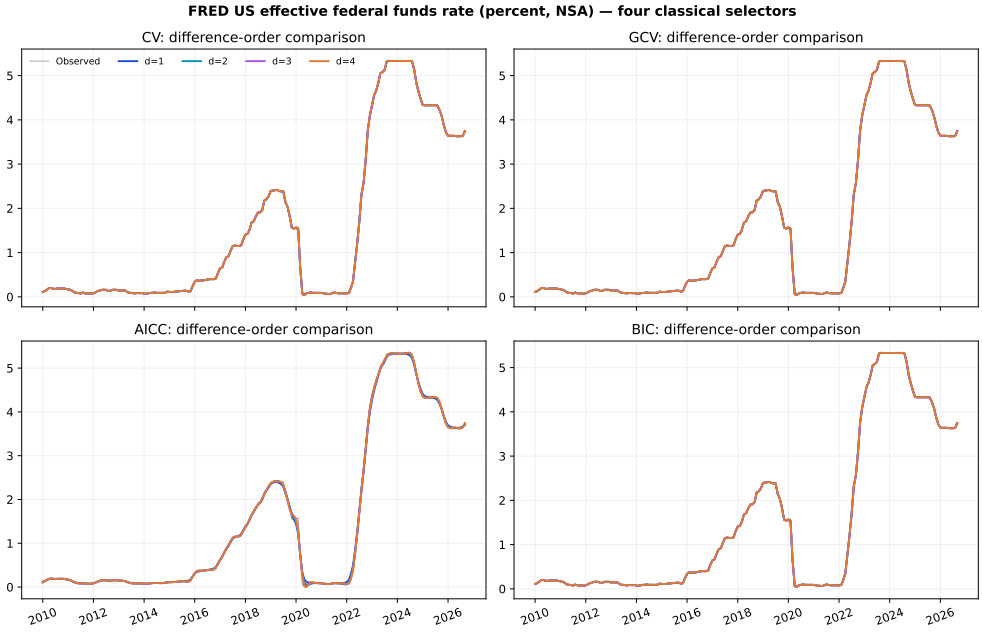


Fig. 7 US monthly not not monthly effective federal funds rate (FEDFUNDS), January 2010–September 2026, with CV, GCV, AICc, and BIC selections across $d = 1, 2, 3, 4$. This independent macroeconomic example illustrates structurally different dynamics from the seasonally varying CO_2 record.

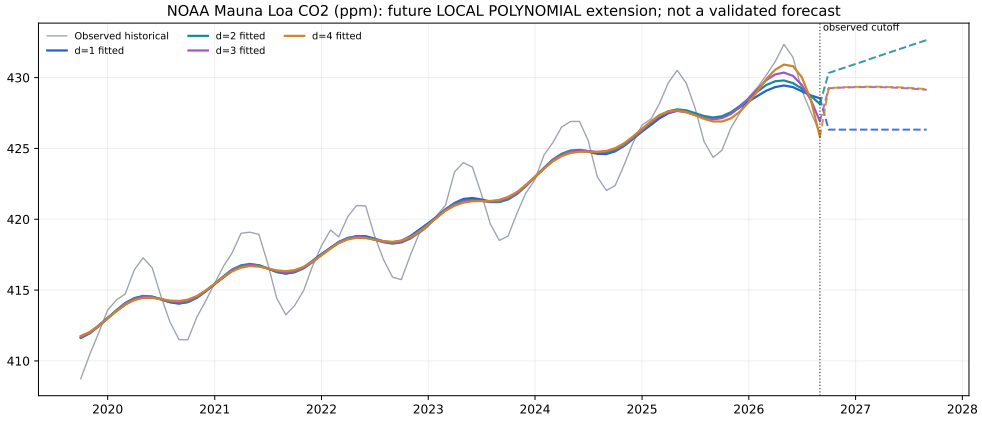


Fig. 8 Descriptive trend extension from all NOAA observations through September 2026 (vertical cutoff) to the subsequent 12 calendar months. Solid lines are fitted trends with common $S = 0.86$; dashed lines are local-polynomial *extrapolations*. They are not observed future data, model-based forecasts, or confidence intervals.

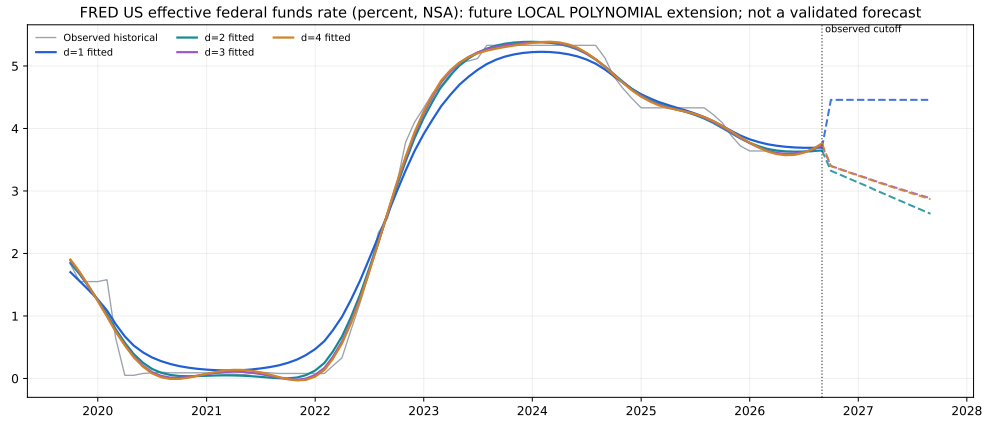


Fig. 9 Corresponding trend-only 12-month extension for not monthly federal funds effective rate. The different orders may imply substantially different extrapolation slopes or curvature even at identical normalized S. Dashed segments are illustrative, not validated forecasts.

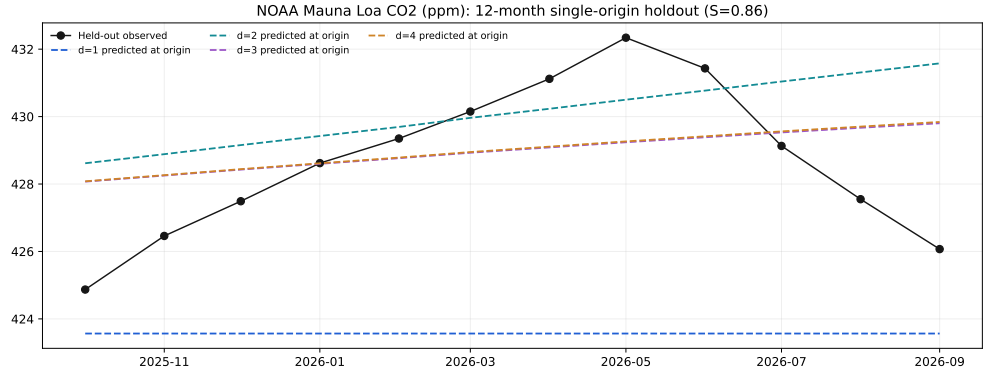


Fig. 10 Historical 12-month holdout observations and trend-only extrapolations made from the preceding training segment, using controlled $S = 0.86$. CO_2 seasonal fluctuations remain in the observations; poor agreement cannot be interpreted as a clean test of trend extraction alone.

criteria selection and several difference orders, paired λ and normalized S reporting, and the sensitivity of trend-only polynomial extrapolations to the penalty order. A single historical holdout is reported as an exploratory diagnostic rather than a validated forecasting comparison. Spectral decomposition makes repeated penalty evaluation transparent and inexpensive after preprocessing; specialized banded solvers remain important competitors for isolated fits and large samples. The source-inspired experiment shows how the same observations can lead to distinct smoothness choices under established criteria, while deterministic checks support correctness in a bounded set of cases. The separate numerical comparison places naive dense normal equations, cached dense solves, reusable SVD, and an independently assembled banded solver under

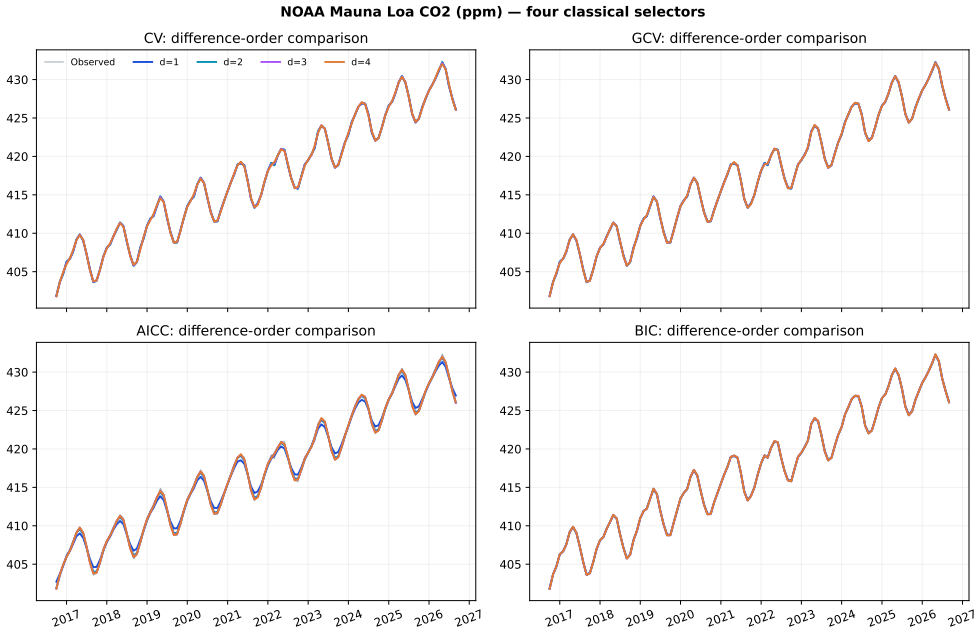


Fig. 11 Sensitivity to the observation window: NOAA Mauna Loa CO_2 over only the most recent 120 published months with the same selection methods and difference orders as Fig. 5. Different windows give different fitted problems and need not yield the same selected normalized smoothness.

matched workloads. This distinction is important: spectral reuse is particularly attractive for repeated trace-based scores, but is not automatically optimal for reconstructing full trends. Broader performance conclusions require reported machine characteristics, workload-specific runtimes, and stability diagnostics. Neither high-order differencing nor likelihood-based smoothing is claimed as a newly discovered concept. No GPU acceleration has been verified in the released CPU implementation.

Declarations

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Competing interests. To be confirmed by the authors before submission.
Ethics approval and consent. Not applicable to the presented numerical algorithms and historical aggregate time-series examples; authors to verify.
Data availability. The simulated data can be reproduced from the supplied scripts and seeds. Current NOAA monthly CO_2 observations and US effective federal funds rate observations from Federal Reserve/FRED are provided as dated, frozen CSV snapshots with source URLs and SHA-256 checksums, along with executable figure-generation code. The archived `statsmodels` 1990–1999 example is retained in the repository but is no longer the submission example.

Code availability. The implementation and paper-specific simulation and benchmark scripts are maintained at https://github.com/heri-espino/trendestimation. A permanently archived release identifier should be added before submission.
Author contributions. To be confirmed and approved by both authors before submission.

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